

Roll No.

TME-501

**B. TECH. (MECHANICAL ENGINEERING)
(FIFTH SEMESTER)**

END SEMESTER EXAMINATION, Dec., 2022

HEAT & MASS TRANSFER

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Derive the 3-D conduction equation in Cartesian coordinate. Also write the special cases. (CO1, 2)

(b) A cylindrical cement tube of radii 0.05 cm and 1.0 cm has wire embedded into it along its axis. To maintain a steady temperature difference of 120°C between the inner and outer surfaces, a current of 5 ampere is made of flow in the wire. Make calculations for the amount of heat generated per metre length and the thermal conductivity of cement.

Take resistance of wire equal to 0.1 ohm per cm of length. (CO1, 2)

P. T. O.

- (c) What is Critical thickness of insulation ? Drive the expression for cylindrical surface. Draw the curve showing relation Heat (Q) and critical radius(r_c). (CO1, 2)
2. (a) What is FINs ? Write the different types of FINs. Drive the heat transfer through short fin with insulated tip. (CO2, 3)
- (b) A long rod 12 mm square section made of low carbon steel protrudes into air at 35°C from a furnace wall at 200°C. The convective heat transfer coefficient is estimated at 22 W/m²K. The conductivity of the material is 51.9 W/mK. Determine the location from the wall at which the temperature will be 60°C. Calculate the temperature at 80 mm from base.
- Calculate the total heat flow and heat convected upto 159 mm and 80 mm lengths. (CO2, 3)
- (c) What is the FIN efficiency and effectiveness ? Why do we use extended surfaces for enhance heat transfer rate ? (CO2, 3)
3. (a) What is convection in heat transfer ? Also the write the importance of dimensional analysis. Explain the difference between Nusselt number and Biot number. (CO2, 3, 4)
- (b) The shell of an experimental boiling water reactor is cylindrical having inside radius 1 m, length 1.25 m and 10 cm wall thickness. The shell is made from alloy steel and a concrete wall 50 cm thick surrounds it. The thermal conductivities of alloy steel and concrete are 22.5 W/m-deg and 1.12 W/m-deg respectively.
- The reactor operates at a power level of 6 kW, of which 4% is lost in heat transfer through the shell. If the inside water is at 150°C, without the interface temperatures (temperatures on the inside and outside of concrete converging). Neglect and resistance to heat flow between water and steel. (CO2, 3, 4)

(3)

- (c) What thermal boundary thickness for fluid flow over a plane surfaces ? Also draw the diagram showing velocity boundary layer thickness and thermal boundary layer thickness. (CO2, 3, 4)
4. (a) What is radiation heat transfer with governing equation ? Also explain the Kirchoff's law and Wien's displacement law. (CO4)
- (b) A furnace having inside temperature of 2250 K has a glass circular viewing of 6 cm diameter. If the Transmissivity of glass is 0.08, make calculations for the heat loss from the glass window due to radiation. (CO4)
- (c) Drive the expression for Radiation heat transfer between two parallel black surfaces. (CO4)
5. (a) What is fouling and scaling in heat exchanger ? Drive the LMTD logarithmic mean temperature difference for counter flow of hot and cold fluid. (CO4, 5)
- (b) What is NTU ? Write the formula of effectiveness of heat exchanger for counter flow and parallel flow having relation with C, NTU. (CO4, 5)
- (c) Determine the area required in parallel flow heat exchanger to cool air from 60°C to 30°C using water available at 20°C. The outlet temperature of the water is 25°C. The overall heat transfer coefficient $U = 320 \text{ W/m}^2\text{K}$. Compare the required for a counter flow exchanger. (CO4, 5)

Roll No.

TME-502

**B. TECH. (ME) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

DESIGN OF MACHINE ELEMENTS—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) Use of Design Data Hand Book is allowed.

(vi) Assume suitable data if required.

1. (a) Define design synthesis. Distinguish between design synthesis and design analysis. What do you understand by size of a product ? Give examples. (CO1)
- (b) Find out the numbers of R20/3 (200,) derived series. It is required to standardise 11 speeds from 72 to 720 r.p.m. for a machine tool. Specify the speeds. (CO1)
- (c) Explain 'Steel' as an Engineering Material. Classify plain carbon steel. Suggest type of material along with its composition and application for 25C12S14, 25Cr4Mo2, X15Cr25Ni12, FeE250. (CO1)

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2. (a) The frame of a hydraulic press consisting of two identical steel plates is shown in Fig. 1. The maximum force P acting on the frame is 20 kN. The plates are made of steel 45C8 with tensile yield strength of 380 N/mm^2 . The factor of safety is 2.5. Determine the plate thickness.

(CO2)

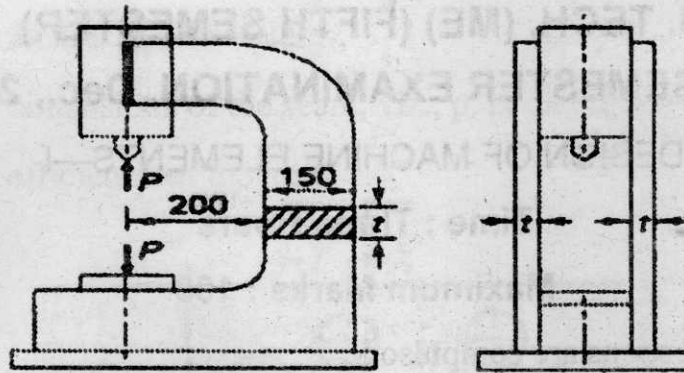


Fig. 1

- (b) The shaft of an overhang crank subjected to a force P of 1 kN is shown in Fig. 2. The shaft is made of plain carbon steel 45C8 and the tensile yield strength is 380 N/mm^2 . The factor of safety is 2. Determine the diameter of the shaft using the maximum shear stress theory. (CO2)

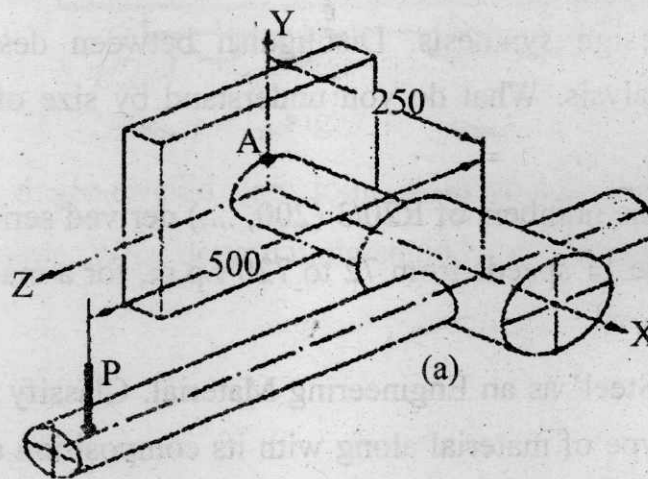


Fig. 2

- (c) Specify the range of magnitude of factor of safety for cast iron components, ductile components, components subjected to fluctuating loads, components surface endurance and buckling respectively. State various failure theories along with their application. (CO2)
3. (a) A flat plate subjected to a tensile force of 5 kN is shown in Fig. 3. The plate material is grey cast iron FG 200 and the factor of safety is 2.5. Determine the thickness of the plate. (CO3)

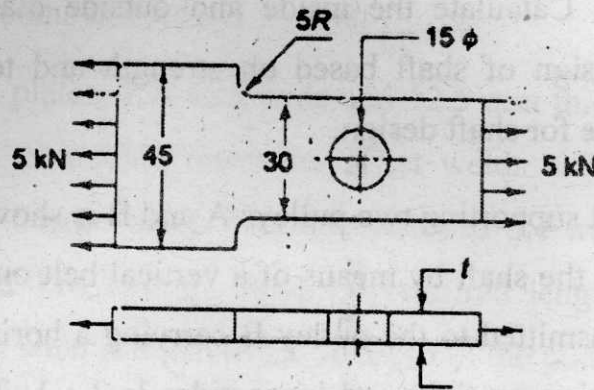


Fig. 3

- (b) A plate made of steel 20C8 ($S_{ut} = 440 \text{ N/mm}^2$) in hot rolled and normalized condition is shown in Fig. 4. It is subjected to a completely reversed axial load of 30 kN. The notch sensitivity factor q can be taken as 0.8 and the expected reliability is 90%. The size factor is 0.85. The factor of safety is 2. Determine the plate thickness for infinite life. (CO3)

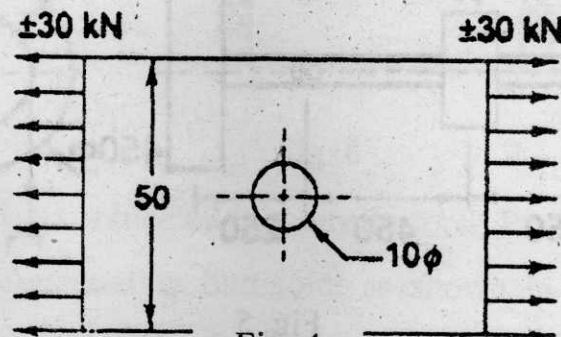


Fig. 4

P. T. O.

- (c) A forged steel bar 50 mm in diameter, is subjected to a reversed bending stress of 250 N/mm^2 . The bar is made of steel 40C8 ($S_{ut} = 600 \text{ N/mm}^2$). Calculate the life of the bar for a reliability of 90%. (CO3)
4. (a) A propeller shaft is required to transmit 45 kW power at 500 r.p.m. It is a hollow shaft, having an inside diameter 0.6 times of outside diameter. It is made of plain carbon steel and the permissible shear stress is 84 N/mm^2 . Calculate the inside and outside diameters of the shaft. Explain design of shaft based on strength and torsional rigidity and ASME code for shaft design. (CO4)
- (b) A line shaft supporting two pulleys A and B is shown in Fig. 5. Power is supplied to the shaft by means of a vertical belt on the pulley A, which is then transmitted to the pulley B carrying a horizontal belt. The ratio of belt tension on tight and loose sides is 3 : 1. The limiting value of tension in the belts is 2.7 kN. The shaft is made of plain carbon steel 40CB ($S_{ut} = 650 \text{ N/mm}^2$ and $S_{yt} = 380 \text{ N/mm}^2$). The pulleys are keyed to the shaft. Determine the diameter of the shaft according to the ASME code if, $k_b = 1.5$ and $k_t = 1.0$. (CO4)

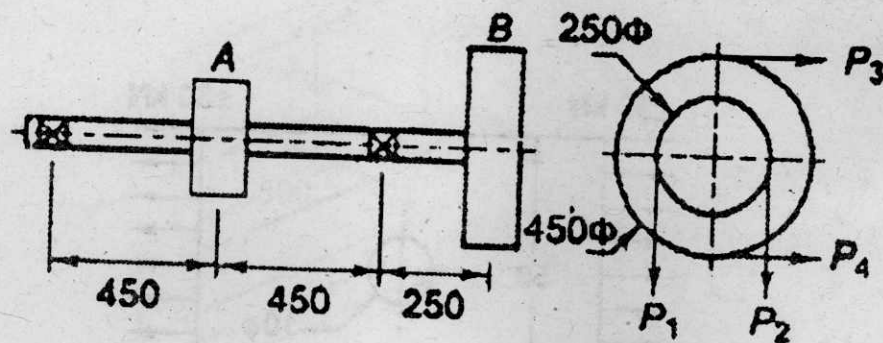


Fig. 5

- (c) It is required to design a square key for fixing a gear on a shaft of 25 mm diameter. The shaft is transmitting 15 kW power at 720 r.p.m. to the gear. The key is made of steel 50C4 ($S_{yt} = 460 \text{ N/mm}^2$) and the factor of safety is 3. For key material, the yield strength in compression can be assumed to be equal to the yield strength in tension. Determine the dimensions of the key. (CO4)
5. (a) Two steel plates, 120 mm wide and 12.5 mm thick, are joined together by means of double transverse fillet welds as shown in Fig. 6. The maximum tensile stress for the plates and the welding material should not exceed 110 N/mm^2 . Find the required length of the weld, if the strength of weld is equal to the strength of the plates. (CO5, 6)

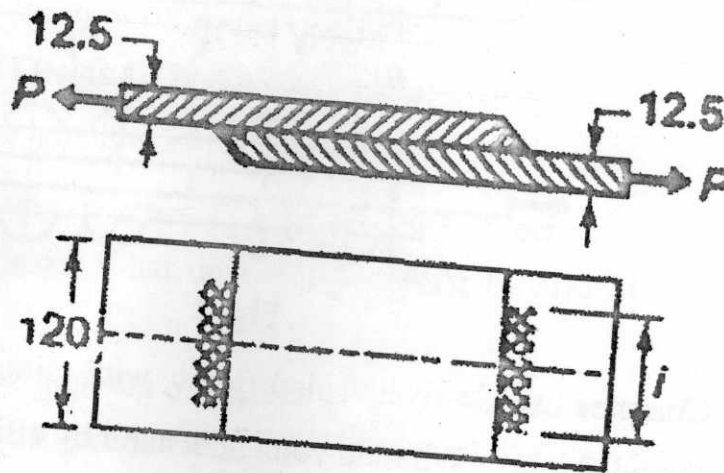


Fig. 6

- (b) Two flat plates subjected to a tensile force P are connected together by means of double-strap butt joint as shown in Fig. 7. The force P is 250 kN and the width of the plate w is 200 mm. The rivets and plates are made of the same steel and the permissible stresses in tension,

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compression and shear are 70, 100 and 60 N/mm² respectively.
Calculate :

(CO5, 6)

- (i) the diameter of the rivets;
- (ii) the thickness of the plates;
- (iii) the dimensions of the seam, viz., p , pt and m ; and
- (iv) the efficiency of the joint.

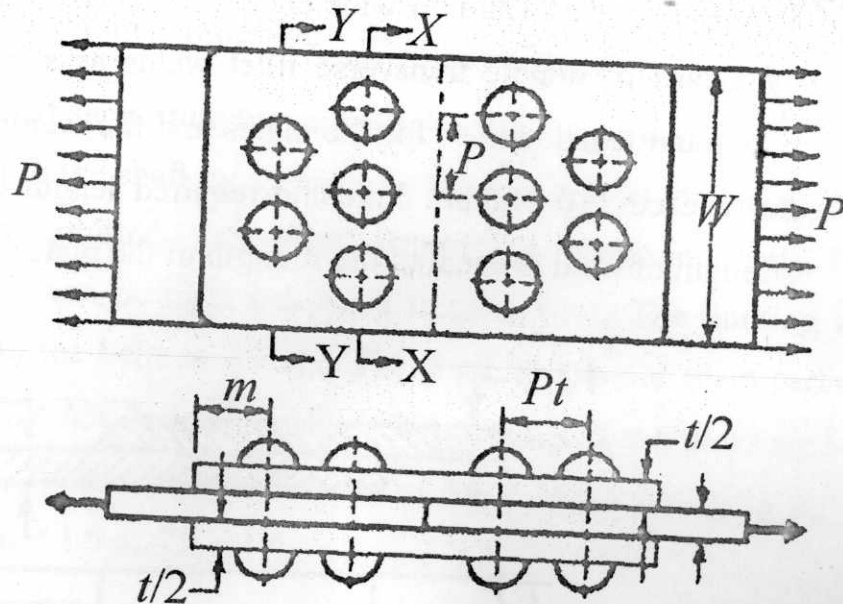


Fig. 7

- (c) Compare single-riveted lap joint with single-strap butt joint and threaded joints. What do you understand by efficiency of riveted joints ?

(CO5, 6)

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TME-503

**B. TECH. (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

DYNAMICS OF MACHINES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the principle of virtual work ? Explain. (CO1, 2)
- (b) Derive an expression for the angular velocity and angular acceleration of the connecting rod. (CO1, 2)
- (c) The crank and connecting rod of a vertical petrol engine, running at 1800 r.p.m. are 60 mm and 270 mm respectively. The diameter of the piston is 100 mm and the mass of the reciprocating parts is 1.2 kg. During the expansion stroke when the crank has turned 20° from the top dead centre, the gas pressure is 650 kN/m^2 . (CO1, 2)

Determine :

- (i) net force on the piston
- (ii) net load on the gudgeon pin
- (iii) thrust on the cylinder walls.

P. T. O.

2. (a) What is meant by cross belt drive ? Find the length of belt in a cross belt drive. (CO3)
- (b) What is the effect of centrifugal tension on tight side and slack side of a belt drive ? Derive the condition for maximum power transmission by a belt drive considering the effect of centrifugal tension. (CO3)
- (c) A casting weighs 6 kN and is freely suspended from a rope which makes 2.5 turns around a drum of 200 mm diameter. If the drum rotates at 40 r.p.m., determine the force required by a man to pull the rope from the other end of the rope. Also find the power to raise the casting. The coefficient of friction is 0.25. (CO3)
3. (a) Explain the method of finding the counter masses in two planes to balance the dynamic unbalance of rotating masses. (CO4)
- (b) A rotor has the following properties : (CO4)

Mass	Magnitude (kg)	Radius of rotation (mm)	Angle (degree)
1	3	30	30
2	4	20	120
3	2	25	270

Angular positions are measured counterclockwise from the reference line along the x-axis. Determine the amount of counter mass at a radial distance of 35 mm required for static balance.

- (c) The axes of three-cylinder air compressor are at 120° to one another and their connecting rods are coupled a single crank. The length of each connecting rod is 240 mm and the stroke is 160 mm. The reciprocating parts have a mass of 2.4 kg per cylinder. Determine the primary and secondary forces if the engine runs at 2000 r.p.m. (CO4)

(3)

4. (a) Sketch a Hartnell governor and deduce a relation to find the stiffness of the spring. (CO5)
- (b) Explain the following terms related to Governor : (CO5)
- (i) Sensitiveness
 - (ii) Hunting
 - (iii) Isochronism
 - (iv) Stability
 - (v) Effort of a Governor
- (c) Each arm of a Porter governor is 200 mm long and is pivoted on the axis of the governor. The radii of rotation of the balls at the minimum and the maximum speeds are 120 mm and 160 mm respectively. The mass of the sleeve is 24 kg and each ball is 4 kg. Find the range of speed of the governor. Also determine the range of speed if the friction at the sleeve is 18 N. (CO5)
5. (a) What are uniform pressure and uniform wear theories related to pivots and collars ? Derive the expression for frictional torque related to flat collar. (CO3, 6, 1)
- (b) What do you mean by gyroscopic couple ? Derive a relation for its magnitude. (CO3, 6, 1)
- (c) What are the conditions for a body to be in equilibrium under the action of two forces, three forces and two forces and torque ? (CO3, 6, 1)

Roll No.

DEEM-01

**B. TECH. (MECHANICAL ENGINEERING)
(FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
ENGINEERING ECONOMICS AND PLANNING**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

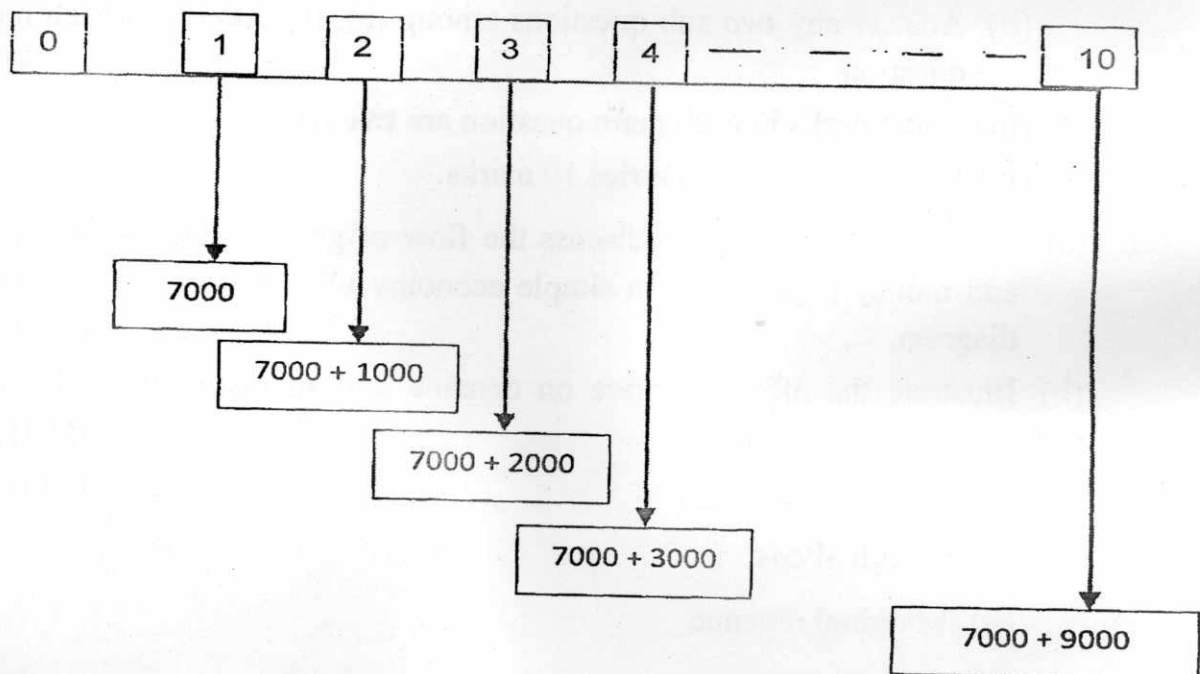
(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define economics. Also discuss the flow of goods, services, resources and money payments in a simple economy with the help of a suitable diagram. (CO1, 2)
- (b) Illustrate the effect of price on demand and supply with the help of diagram. (CO1, 2)
- (c) Explain the following : (CO1, 2)
 - (i) Marginal cost
 - (ii) Marginal revenue
 - (iii) Average cost
 - (iv) Sunk cost

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2. (a) Briefly explain the Engineering Economics. What do you mean by Microeconomics and Macroeconomics ? (CO1, 4)
- (b) The automobile company recently advertised its car for a down payment of ₹ 1,50,000. Alternatively, the car can be taken home by customers without making any payment, but they have to pay an equal yearly amount of ₹ 25,000 for 15 years at an interest rate of 10%, compounded annually. You are asked to advice the best alternative for the customer based on the present worth method of comparison. (CO1, 4)
- (c) Consider the following cash flow diagram. Find the present worth using an interest rate of 15% compounded annually. (CO1, 4)



3. (a) Explain the law of returns, giving reasons for increasing returns, constant returns and decreasing returns. (CO2, 4)

- (b) You are the manager of a store that carries generic soft drinks. Due to a local economic boom, your customer's incomes are forecasted to rise by 5% during the next month. The income elasticity of demand for these products is estimated to be -2.0 . Estimate the change in the quantity of your soft drink orders required to accommodate the new demand without a surplus or shortage of inventory. (CO2, 4)
- (c) Suppose that demand and supply of apples are described by the following equations : (CO1, CO2)
- $P = 100 - 3Q$ (demand)
- $P = 20 + Q$ (supply)
- (i) Calculate the equilibrium price and quantity.
- (ii) Suppose a \$4 tax is placed on apples. What is the new equilibrium quantity ?
- How much do consumers pay to get this quantity ? How much do suppliers receive for selling the quantity ?
4. (a) Explain flow in economy. What do you mean by circular economy ? (CO2, 3, 4)
- (b) A man owns a corner plot. He must decide which of the several alternatives to select in trying to obtain a desirable return on his investment. After much study and calculation, he decides that the best alternatives are as given in the following table : (CO2, 3, 4)

	Build gas station	Build soft ice-cream stand
Fixed cost (₹)	20,00,000	36,00,000
Annual property taxes (₹)	80,000	1,50,000
Annual income (₹)	8,00,000	9,80,000
Life of building (Years)	20	20
Salvage value (₹)	0	0

- (c) A person is planning for his retired life. He has 15 years of service, he would like to deposit 20% of salary, which is ₹ 12,000 at the end of 1st year and there after he wishes to increase his deposit by ₹ 2,000 more every year along with ₹ 12,000 for next 14 years. What will be the maturity amount of this deposit, if the interest rates are 10% and 12% per year ? (CO2, 3, 4)

5. (a) Explain the time value of money. Write various interest formulas.

(CO5, 6)

- (b) Krishna Company Ltd. has following details in Rupees :

(CO5, 6)

Fixed cost = 40,00,000

Variable cost per unit = 300

Selling price per unit = 500

Find :

- (i) The break-even sales quantity
 - (ii) The break-even sales
 - (iii) If the actual production quantity is 1,20,000, calculate contribution and margin by safety.
- (c) The following table gives the operation cost, maintenance cost and salvage value at the end of every year of a machine whose purchase value is ₹ 20,000. (CO5, 6)

(5)

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Find the economic life of the machine assuming interest rate, $i = 15\%$.

End of Year (n)	Operation cost at the end of year (₹)	Maintenance cost at the end of year (₹)	Salvage value at the end of year (₹)
1	3,000	300	9,000
2	4,000	400	8,000
3	5,000	500	7,000
4	6,000	600	6,000
5	7,000	700	5,000
6	8,000	800	4,000
7	9,000	900	3,000
8	10,000	1,000	2,000
9	11,000	1,100	1,000
10	12,000	1,200	0

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DEMM-02

B. TECH. (ME) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
RAPID PROTOTYPING AND TOOLING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How can one classify tool paths for all RP processes into different categories ? (CO2)
- (b) Explain the different types of STL file problems. (CO2)
- (c) Write the difference between Hard Tooling and Soft tooling. (CO6)
2. (a) What are the different steps in a typical rapid prototyping process ? (CO1)
- (b) What is Rapid Tooling ? What are the types of tooling ? Compare Rapid and Conventional tooling. (CO6)
- (c) What is Indirect Rapid Tooling ? Write a short notes on silicon rubber tooling. (CO6)

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(2)

3. (a) Explain the process of facet generation for gaps by taking an example with odd and even number of edges. (CO4)
(b) Write down the steps for solving the "Missing Facets" problem in STL file. (CO4)
(c) What are the differences between traditional and rapid prototyping ? (CO1)
4. (a) What are the different accuracy problems in SLA process ? (CO4)
(b) What is Rapid Prototyping Process ? What are the *three* types of fundamental fabrication processes ? (CO1)
(c) Explain the initial graphics exchange specification and computerized tomography format. (CO2)
5. (a) Explain Electron Beam Melting (EBM) process with its advantages and disadvantages. (CO3, CO5)
(b) Explain Stereolithography (SLA) process with its advantages and disadvantages. (CO3, CO5)
(c) Explain Fused Deposition Modelling (FDM) process with its advantages and disadvantages. (CO3, CO5)

Roll No.

DETF-01

B. TECH. (ME) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022

SOLAR ENERGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain different solar intensity measuring instruments with a neat sketch. (CO1, CO2)
- (b) Explain different sun earth angles. (CO1, CO2)
- (c) Explain concentrator collector with a neat sketch. (CO1, CO2)
2. (a) Explain different performance parameters of solar collector. Also, explain the sensitivity analysis of solar collector. (CO2, CO3)
- (b) Write energy balance equations for flat plate collector and hence find the expression of temperature at the outlet of flat plate collector. (CO2, CO3)

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- (c) Explain the working of single slope passive solar still with neat sketch.
Also, write energy balance equations for single slope passive solar still.

(CO2, CO3)

3. (a) Explain the working principle of double slope solar still with a neat sketch. Also, write energy balance equation for double slope solar still.

(CO3)

- (b) Explain the different performance parameters of solar still. (CO3)

- (c) Explain the working principle of conical solar still and tubular solar still with a neat sketch. (CO3)

4. (a) Explain the working principle of a typical solar dryer with a neat sketch.
Also, write the application of solar dryer. (CO4, CO5)

- (b) Explain a typical solar house with a neat sketch. Also, write the various applications of solar house. (CO4, CO5)

- (c) Explain the different methods of sensible heat storage with a neat sketch. (CO4, CO5)

5. (a) Explain the characteristic curves of solar panel with a neat sketch.

(CO5, CO6)

- (b) What do you understand by capital recovery factor and sinking fund factor? Derive the expression for uniform annual cost. (CO5, CO6)

(3)

- (c) Two solar water heating systems have the following cost comparison.
Which system is more economical if the money is worth 10 percent per year : (CO5, CO6)

Cost components	System (A)	System (B)
First cost	₹ 20,000	30,000
Uniform end-of-year maintenance	₹ 4,000	3,000
Salvage value	₹ 500	1,500
Service life (years)	2	3

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DETF-02

**B. TECH. (ME) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec.,-2022**

POWER PLANT ENGINEERING

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) What is doubling time in power plant engineering ? How is the load-duration curve constructed ? (CO1)

(b) What do you understand by depreciation ? How is the depreciation rate determined ? (CO1)

(c) A power plant has the following annual factors : (CO1)

Load factor = 75%, capacity factor = 60%, use factor = 65%, maximum demand = 60 MW.

Estimate :

- (i) Annual energy production
(ii) Reserve capacity over and above the peak load

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2. (a) What factors should be taken into consideration while selecting the site for steam power plant ? (CO2)
- (b) Define the chimney efficiency and find out expression for the same. (CO2)
- (c) Steam with absolute velocity of 300 m/s is supplied through a nozzle to a single stage impulse turbine. The nozzle angle is 25° . The mean diameter of the blade rotor is 1 meter and it has a speed of 2000 r.p.m. Find suitable blade angles for zero axial thrust. If the blade velocity coefficient is 0.9 and the steam flow rate is 10 kg/s, calculate the power developed. (CO2)
3. (a) Explain the essential components of a Diesel Power Plant. (CO3, 4)
- (b) Derive the optimum pressure ratio in an ideal gas turbine for maximum work output. (CO3, 4)
- (c) Draw the layout of Gas Power Plant. List the advantages and limitations of gas power plant. (CO3, 4)
4. (a) State the essential elements of Hydroelectric Power Plant ? Explain the different methods of classifying a Hydroelectric Power Plant. (CO5)
- (b) Explain with neat sketch. Explain the construction and working of Francis Turbine Turbine. (CO5)
- (c) A pelton wheel has a mean bucket speed of 10 m/s with a jet of water flowing at a rate of $0.7 \text{ m}^3/\text{s}$ under a head of 30 m. The buckets deflect the jet through an angle of 160° . Calculate the power and efficiency of the turbine. Take $C_v = 0.98$ (CO5)

(3)

5. (a) Write short notes on Boiling Water Reactor (BWR) and Pressurized Water Reactor (PWR). (CO6)
- (b) Calculate the mass defect and binding energy per nucleon (MeV) of oxygen. Given : $m_p = 1.007277$ amu, $m_n = 1.0086665$ amu, $m_e = 0.00055$ amu, atomic mass of oxygen = 15.99491 amu. (CO6)
- (c) What is CANDU-type reactor ? Explain with neat diagram. (CO6)

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DETF-05

**B. TECH. (ME) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
INTERNAL COMBUSTION ENGINE**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the working of a four stroke SI engine with the help of a neat sketch. (CO1)
- (b) Explain the various parameters used to evaluate the engine performance. (CO1)
- (c) Explain the effects of following variables on fuel-air cycle : (CO5)
 - (i) composition of cylinder gases
 - (ii) variable specific heats
 - (iii) dissociation
2. (a) What do you mean by carburetion ? Explain the air-fuel ratio requirement of a petrol engine from no load to full load. (CO2)
- (b) With a neat sketch explain the various components of a fuel feed system of a CI engine. (CO2)

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(2)

- (c) Explain the following systems of a simple carburetor : (CO2)
- (i) main metering system
 - (ii) idling system
 - (iii) choke and
 - (iv) throttle
3. (a) Explain the stages of combustion in CI engines with the help of a diagram. (CO3)
- (b) What do you mean by abnormal combustion ? Explain the phenomenon of knock in SI engine. (CO3)
- (c) Explain various factors affecting the ignition lag. (CO3)
4. (a) What are the emissions that come out from the engine exhaust ? Describe the causes of hydrocarbon emissions from SI and CI engines. (CO4)
- (b) What is the need of exhaust treatment devices for engines ? Explain how a catalytic converter controls the exhaust emission. (CO4)
- (c) What is EGR ? Explain how it reduces the NO_x emission ? (CO4)
5. (a) What do you mean by supercharging and turbocharging ? Explain the principle of supercharging. (CO6)
- (b) Explain the concept of Low Temperature Combustion (LTC). Also, explain the HCCI and RCCI types of LTC. (CO6)
- (c) Explain the effect of spark timing, mixture composition, compression ratio, load and speed on SI engine performance. (CO5)

Roll No.

TAS-501

**B. TECH. (ME) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

AERODYNAMICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the various sources of drag on an aeroplane ? Explain how they are caused and method to reduce them. (CO1)

(b) State the difference between one, two and three dimensional flows. (CO2)

(c) A plane is in level flight at constant speed and each of its two wings has an area of 25 m^2 . If the speed of the air on the upper and lower surfaces of the wing are 270 km h^{-1} and 234 km h^{-1} respectively, determine the mass of the plane. (CO1)

P. T. O.

2. (a) Explain NACA 4 digit series Airfoil configuration , and explain NACA 2412 and NACA 0020. (CO2)
- (b) Define Kutta condition and show with a neat sketch the effect of stagnation points when : (CO2)
- (i) $\Gamma < 4\pi V_\alpha R$
 - (ii) $\Gamma = 4\pi V_\alpha R$
 - (iii) $\Gamma > 4\pi V_\alpha R$
- (c) Draw a typical CL vs alpha curve for a unsymmetrical airfoil and show the lift curve slope , maximum lift, stalling angle and show flow pattern at stalling angle. (CO2)
3. (a) How is the aerodynamics over a wing different from that over an airfoil ? (CO3)
- (b) Explain the phenomena of downwash and the induced drag. (CO3)
- (c) Explain with a neat sketch the formation of starting vortex ? (CO3)
4. (a) Define the Critical Mach number and show different regimes of flow : (CO4)
- (i) Subsonic
 - (ii) Transonic
 - (iii) Supersonic
 - (iv) Hypersonic
- (b) With a neat sketch show that drag is constant till critical Mach number and rises because of compressibility and also show what

(3)

happens

at

$M = 1$. Explain the complete process in detail.

(CO4)

(CO6)

(c) Explain of the following :

(i) The Area rule

(ii) Supercritical Airfoil

5. (a) What is the difference between subsonic and supersonic airflow ?
(CO5)

(b) Discuss the Aerodynamics Phenomenon involved in formation flying.
(CO5)

(c) Explain and show with a neat sketch the effects of viscosity on shear stress ?
(CO5)

Roll No.

TRA-501

**B. TECH. (ME-RA) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

ROBOTICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

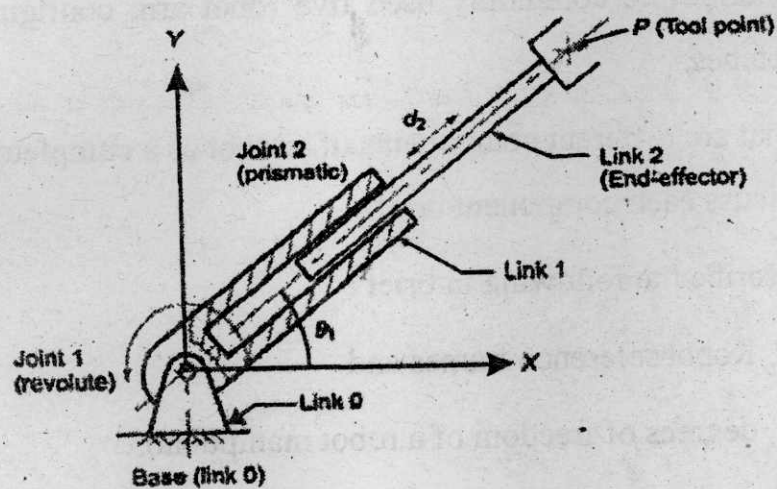
(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

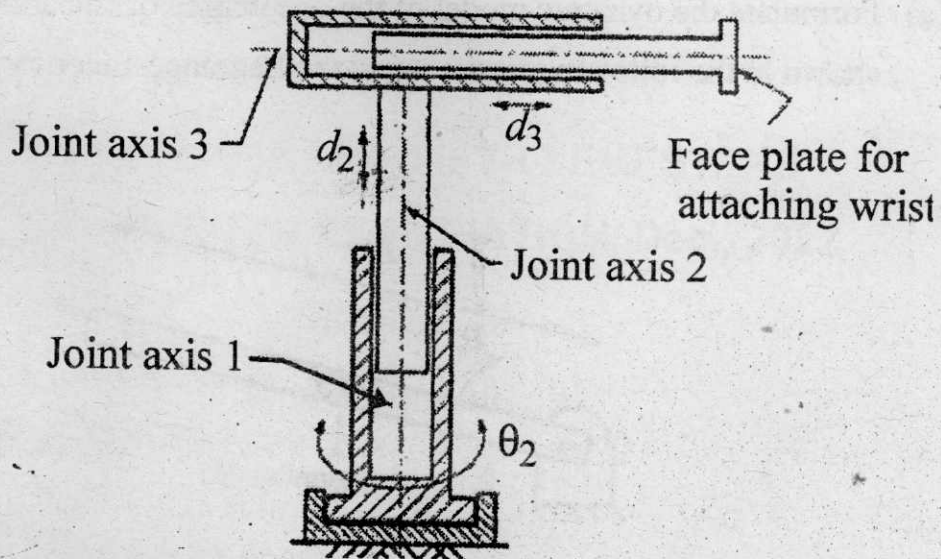
1. (a) Describe the commonly used five robot arm configurations with neat sketches. (CO1)
- (b) What are different components of a robot as a complete robotic system ? Discuss each component briefly. (CO1)
- (c) Describe the following in brief : (CO1)
 - (i) Robot reference frames and
 - (ii) degrees of freedom of a robot manipulator.

P. T. O.

2. (a) Describe the mapping in general frames with neat sketch and also discuss about the homogenous transform. (CO2)
- (b) A frame $\{V\}$ is rotated relative to a frame $\{U\}$ about z-axis by 30 degree, translated 10 units in x and translated 6 units in y; the z-axis is pointing out of the page. Position vector of a point with respect to frame $\{V\}$ is given by $[1, 3, 0]^T$. Calculate the position of that point with respect to frame $\{U\}$. (CO2)
- (c) A frame $\{B\}$ is rotated relative to another frame $\{A\}$ about Z axis by 45 degrees and translated 2 units in X and 3 units in Y. The Z axis is pointing out of the page. Thus, we have a description of ${}^A_B T$. Find ${}^B_A T$. (CO2)
3. (a) What do you understand by the forward and inverse kinematics of manipulators ? Also, discuss about the joint-link parameters of robot kinematics. (CO3)
- (b) Formulate the inverse kinematic model for the 2-DoF, RP planar manipulator shown in the figure below : (CO3)

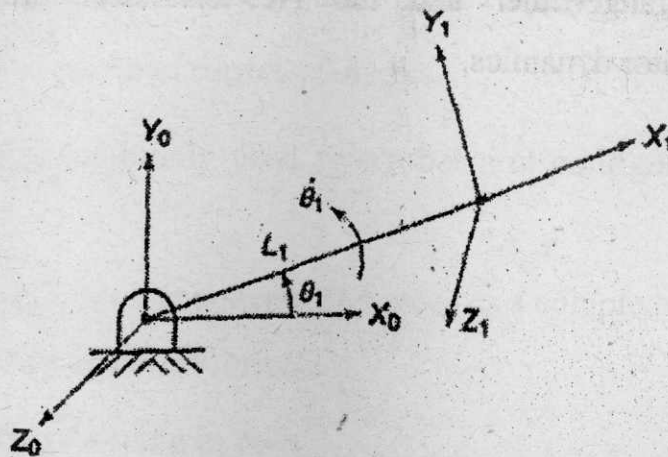


- (c) Formulate the forward kinematic model for the 3-DoF, RPP manipulator shown in the figure below : (CO3)



4. (a) Calculate the linear and angular velocities of the end-effector for the one degree of freedom manipulator shown in the following figure.

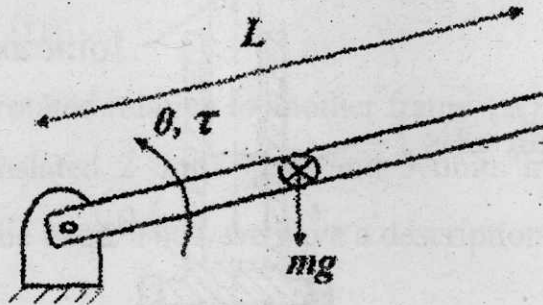
(CO4, 5)



- (b) What do you understand by the manipulator differential motion ? Also, describe the relationship between Jacobian matrix, joint velocities vector and the end-effector velocities vector. (CO4, 5)

P. T. O.

- (c) Discuss about the Jacobian inverse and the Static force analysis for robot manipulators. (CO4, 5)
5. (a) Formulate the dynamic model of the one degree of freedom manipulator shown in the following figure using the Lagrange-Euler formulation. (CO6)



- (b) Formulate the dynamic model of the one degree of freedom manipulator shown in the figure above using the Newton-Euler formulation. (CO6)
- (c) What do you understand by the manipulator dynamics? Discuss about the Lagrange-Euler and the Newton-Euler formulations for the manipulator dynamics. (CO6)

60

Roll No.

TAE-501

**B. TECH. (ME) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

ELECTRIC VEHICLE TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the factors that have propelled the use of Electric vehicles.

(CO1)

(b) Consider an electric vehicle with a speed of 80 km/hr. with a coefficient of drag of 0.18, with a frontal area of 2.7 sq. m. If the energy available is 20 kWhr and the range without any wind is 140 km, find the range with wind speed of 40 km/hr. Take the density of air to be 1.25 kg/cu m.

(CO1)

(c) Compare the merits and demerits of different configurations of electric vehicles.

(CO2)

2. (a) Explain the factors influencing battery life.

(CO2)

P. T. O.

(2)

- (b) Compare the different types of batteries and suggest their suitability for Electric vehicles. (CO3)
- (c) Explain the construction and working of the PEMFC. (CO3)
- 3. (a) Write about the requirements of motor drives for electric vehicles. (CO4)
- (b) Describe the construction and working of switched reluctance motor drive for electric vehicles. (CO4)
- (c) Compare the different DC motor drives used in electric propulsion. (CO4)
- 4. (a) Construct a diagram showing the various components in a series hybrid electric drive train. (CO5)
- (b) Sketch a flow chart describing the control strategy of a parallel hybrid electric vehicle. (CO5)
- (c) Describe the different control strategies of series hybrid electric vehicle. (CO5)
- 5. (a) Explain the construction and working of the DC-DC converter. (CO6)
- (b) Compare the different charging levels of AC and DC charging. (CO6)
- (c) Evaluate the merits and demerits of different types of chargers. (CO6)

Roll No.

TMA-502

B. TECH. (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

COMPUTER BASED NUMERICAL AND STATISTICAL TECHNIQUE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each question carries 10 marks.

1. (a) (i) What do you mean by truncation error ? Explain with examples.

(ii) $\sqrt{10} = 3.162$ and $e = 2.718$ correct to three decimal places. Find the percentage error in their difference. (CO1)

(b) Obtain the solution of the following system using Gauss-Seidel iteration method : (CO1)

$$4x_1 - 5x_2 + x_3 = 2$$

$$3x_1 + x_2 - 2x_3 = 9$$

$$x_1 + 4x_2 + x_3 = 5.$$

P. T. O.

- (c) Find the real root of the $2x - \log_{10} x = 7$ correct to four decimal places using the iterative method. Also, compare this result obtained by Newton-Raphson's method. (CO1)

2. (a) From the following table, find the number of students who obtain less than 45 marks : (CO2)

Marks	No. of Students
30—40	31
40—50	42
50—60	51
60—70	35
70—80	31
80—90	10

- (b) (i) Obtain the function whose first difference is $2x^3 - 3x^2 + 3x - 10 = 0$.

- (ii) By mean of the Newton's divided difference formula, find $f(8)$ from the following table : (CO2)

X	$f(x)$
4	48
5	100
7	294
10	900
11	1210
13	2028

(c) Use Stirling's formula to find $f(28)$ from the following table :

(CO2)

X	$f(x)$
20	49225
25	48316
30	47236
35	45926
40	44306

3. (a) Find the value of $\log_e 3$ from $\int_0^1 \frac{x^3}{1+x^3} dx$ using Simpson's $1/3^{\text{rd}}$ rule by dividing the range into 6 equal parts. (CO3)

(b) Explain Newton-Cote's formula. Using Weddle's rule obtain the value of the integration $\int_0^5 \frac{1}{4x+5} dx$. (CO3)

(c) The following table gives the velocity of a train which starts from rest, time being reckoned in minutes from the start and speed in kilometer per hour : (CO3)

Minutes	Speed (km/hr)
0	0
2	10
4	18
6	25

P. T. O.

8	29
10	32
12	20
14	11
16	5
18	2
20	0

Estimate the total distance in 20 minutes.

4. (a) Employ Picard's method to find $y(0.2)$ & $y(0.4)$ given that : (CO4)

$$\frac{dy}{dx} = x(1 + x^3y), y(0) = 3$$

- (b) Using the Runge-Kutta method of fourth order, solve the following differential equation : (CO4)

$$\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}, y(0) = 1 \text{ at } x = 0.2 \text{ \& } 0.4.$$

- (c) Using Milne's method, solve

$$\frac{dy}{dx} = 1 + y^2, y(0) = 0$$

$$y(0.2) = 0.2027$$

$$y(0.4) = 0.4228$$

and

$$y(0.6) = 0.6841$$

Obtain $y(0.8)$ & $y(0.1)$.

(CO4)

(5)

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5. (a) Fit a curve of the form $y = ax^b$ to the data given below :

(CO5)

X	Y
1	7.1
2	27.8
3	62.1
4	110
5	161

- (b) Obtain the cubic spline fit for the following data :

(CO5)

X	$f(x)$
1	1
2	5
3	11
4	8

Using the end conditions $y''(1) = 0 = y''(4)$. Hence compute $y(1.5)$.

- (c) (i) If θ is the acute angle between the two regression lines in the case of two variables x and y , show that :

(CO5)

$$\tan \theta = \frac{1 - r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

where each notation has its usual meaning.

P. T. O.

- (ii) For 10 observations on price (x) and supply (y), the following data were obtained (in appropriate units) :

$$\Sigma x = 130$$

$$\Sigma y = 220$$

$$\Sigma x^2 = 2288$$

$$\Sigma y^2 = 5506$$

$$\Sigma xy = 3467$$

Obtain the two lines of regression and estimate the supply when the price is 16 units.

(CO5)